

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (EE) Examination

Dec 2012

EE 705 Modern Power System Operations

Date: 22.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1 Draw the flowchart of Newton-Raphson (NR) method for load flow analysis and show all the computational procedure. Consider the presence of all buses. 07
- Q – 2 (a) Derive the linearized models of the following: 07
- (a) Governor Model
 - (b) Tie-line Model
- (b) What is DISCO Participation Matrix (DPM)? What is the significance of Contract Participation Factors (cpfs). Draw and explain the two-area AGC block diagram in restructured power system. 07

OR

- Q – 2 (a) Explain lambda iteration method for the economic dispatch problem considering thermal system dispatching with network losses included 07
- (b) Explain Gradient Methods of economic operation of power system 07
- Q – 3 Attempt any two 14
- (1) Discuss the different constrain for unit commitment
 - (2) Write a short note on dynamic programming
 - (3) What is demand side management? Explain utility load shape objectives and demand side management alternatives.

SECTION – II

- Q – 4 Derive Maximum Likely-hood Weighted Least Squares Estimation method for state estimation in power system 07
- Q – 5 (a) Explain the orthogonal decomposition in state estimation 07
- (b) Explain the methods for the bad data detection and measurement 07

OR

Q – 5 (a) Derive the linear sensitivity factors for power system security (i) Generation shift factor and (ii) line outage distribution factor 07

(b) Explain the following w.r.t. to power system security (a) Performance Index (b) Bounding. 07

Q – 6 Attempt any two 14

- (1) Priority list methods for unit commitment
- (2) Write a short note on ancillary services
- (3) Application of state estimation with schematic diagram
